Volumetric input

S slices (3D imaging)

Frozen visual interface

visual encoder

Pretrained Weights

Pretrained ViT

Visual tokens

N tokens

What abnormality is present ?

MedPruner++

Relational feature composition

Compact visual tokens

VLM for downstream task

text decoder

Pretrained Weights

VQA

Report

…

Content composition

Ordered support construction

$g_{m,p}=\frac{1}{|I_m|}\sum_{s\in I_m} x_{s,p}$

Allocate unit budgets across support units

Relational allocation

$\pi_m=\frac{exp(\eta_m)}{\sum_n exp(\eta_n)}, K_m=[\pi_m B]$

Select distinctive Top-K anchors

Anchor Neighbor Selected

Radial-Angular Selection

$S_m = TopK_p(u^H_{m,p}, K_m)$

Anchors (S_m)

Mean

Neighbor

Refined anchors $(\tilde{x}_{m,p})$

Ordered tokens (Keep source order)

Local refinement & order

$\tilde{x}_{m,p} = \frac{1}{2}(g_{m,p} + \frac{\sum_{q\in R_{m,p}} g_{m,p}}{\max(1, |R_{m,p}|)})$

Capacity control

Model-level attention distribution

Focused attention (low entropy)

Dispersed attention (high entropy)

Determine overall budget

Manual

$\rho \in \{10\%, 20\%, 30\%\}$

Adaptive

CLIP

$B = [\rho N]$

Normalized entropy $\bar{H}$

$\rho_t = clip\left(\frac{\bar{H}^3}{2}; 0, \rho_{max}\right)$

Provide budget B to content composition